This assignment will not be graded and is only for practice.

Level 1

1) If $(1, 2, 3) + v = (3, 2, 1)$ in the vector space R^3 with usual addition and scalar multiplication, where $v = (a, b, c)$, then find the value of $a + b + c$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

2) Which of the following statements are related to the cancellation law for vectors x, y, z in a vector space.

Statement P: $x + y = z + y \implies x = z.$
Statement Q: $x + y = z + y \implies x = y.$
Statement R: $x + y = z + y \implies y = z.$
Statement S: $x + y + z = x \implies y + z = 0$

Find the number of correct statements

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

3) Which of the following options is/are true for a non zero vector space V? ***1 point***

- ☐ $x + y = z + y \implies x = z \ \forall x, y, z \in V.$
- ☐ $ax = bx, \ \forall\, x \in V \implies a = b, \ \text{where } a, b \in \mathrm{R}.$
- ☐ $ax = ay, \ \forall\, a \in \mathrm{R} \implies x = y, \ \text{where } x, y \in V.$

☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x + y = z + y \implies x = z \; \forall x, y, z \in V$.
$ax = bx, \; \forall \, x \in V \implies a = b, \;$ where $a, b \in \mathrm{R}$.
$ax = ay, \; \forall \, a \in \mathrm{R} \implies x = y, \;$ where $x, y \in V$.

4) Which of the following options is/are true for a non zero vector space V? ***1 point***

☐ $av = 0, \; \forall \, v \in V \implies a = 0.$

☐ $av = 0, \; \forall \, a \in \mathrm{R} \implies v = 0.$

☐ $c \in \mathrm{R}$ and $v, w \in V$ such that, $c(v - w) = cv - cw.$

☐ $c((v + w) - (v - w) - 2w) = 0 \; \forall c \in \mathrm{R}, \;$ and $v, w \in V$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$av = 0, \; \forall \, v \in V \implies a = 0.$
$av = 0, \; \forall \, a \in \mathrm{R} \implies v = 0.$
$c((v + w) - (v - w) - 2w) = 0 \; \forall c \in \mathrm{R}, \;$ and $v, w \in V$

5) Let $(3, 2, -3) \in \mathrm{R}^3$ be a vector from the vector space R^3 with usual addition and scalar multiplication. If $c(3, 2, -3) = (-6, -4, 6)$, find the value of c.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -2

1 point

6) Which of the following options is/are true? ***1 point***

- [ ] The zero vector is a unique vector in a vector space.
- [ ] Inverse of a vector is a unique vector in a vector space.
- [ ] $0v = 0$, where v is a vector from a vector space of R.
- [ ] $c0 = 0$, where $c \in \mathrm{R}$ and 0 is the zero vector from a vector space over R.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The zero vector is a unique vector in a vector space.
Inverse of a vector is a unique vector in a vector space.
$c0 = 0$, where $c \in \mathrm{R}$ and 0 is the zero vector from a vector space over R.

Level 2

Let V be a plane plane parallel to the XY-plane. Any plane parallel to XY-plane is given by $z = c$. We define addition of two vectors $v_1 = (x_1, y_1, c)$ and $v_2 = (x_2, y_2, c)$ on V as follows: First project v_1 and v_2 on the XY-plane (we will get the vectors $(x_1, y_1, 0)$ and $(x_2, y_2, 0)$ by projection on the XY-plane) and then calculate the addition of the vectors we obtained by the projection on the XY-plane (we will obtain $(x_1 + x_2, y_1 + y_2, 0)$). Then project the obtained vector back to the plane V (we will obtain the vector $(x_1 + x_2, y_1 + y_2, c)$).
Let V be a plane which is parallel to the XY-plane, defined by $z = 2$. Let $v_1 = (1, 2, 2)$ and $v_2 = (0, 3, 2)$ be in V. Answer questions 7 and 8.

7) If $v_1' = (a_1, b_1, c_1)$ and $v_2' = (a_2, b_2, c_2)$ are the projections of v_1 and v_2 on the XY - plane, respectively, then find the value of $2(a_1 + b_1 + c_1) - (a_2 + b_2 + c_2)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

8) If $v_1 + v_2 = (a_1, b_1, c_1)$ as per the addition defined above, then find the value of $(2a_1 + b_1 - 2c_1)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Let V be a plane plane parallel to the XY-plane. Any plane parallel to the XY-plane is given by $z = c$. We define scalar multiplication of a vector $v = (x, y, c)$ on V as follows: First project v on the XY-plane first (we will get the vector $(x, y, 0)$ by projection on the XY-plane) and then calculate the scalar (α) multiple of the vector we obtained by the projection on the XY-plane (we will obtain $(\alpha x, \alpha y, 0)$). Then project the obtained vector back to the plane V (we will obtain the vector $(\alpha x, \alpha y, c)$).
Let V be a plane which is parallel to the XY-plane, defined by $z = 2$. Let $v = (1, 2, 2)$ be in V. Answer questions 9 and 10.

9) If $v' = (a_1, b_1, c_1)$ is the projection of v on the XY - plane, then find the value of $(a_1 + 2b_1 + c_1)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

10) If $4v = (a_1, b_1, c_1)$ as per the scalar multiplication defined above, then find the value of $(a_1 + 2b_1 + c_1)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 22

1 point

Your score is: 0/10